

RIGIDITY AND UNIVERSAL CONTRACTIONS OF HARD-SPHERE KINETIC COTANGENTS

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ABSTRACT. A phase is introduced as a genuine positive latent variable with conditional microscopic path laws. The joint labelled experiment has a good large-deviation principle and contracts to the minimum of the phase costs and component rates. Restricted pressures use one consistent sign and the exact finite-volume value is a log-sum-exp. Dynamic consistency is restored by carrying the adapted phase posterior in the state; the marginal log-sum alone is not called a semigroup. Quantitative complex dominance gives phase-local response charts, and second-order component weights determine labelled Gaussian mixtures at coexistence.

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1. ROUND-TWENTY-ONE POSITIVE CLOSURE: GENUINE FINITE-VOLUME PHASES, SHARED-POLICY POSTERIOR CONTROL, AND CORRECTED COEXISTENCE LIMITS

This section replaces the former Round-Seventeen source. A phase is an actual finite-volume latent label defined by a measurable partition of the original microscopic experiment. The sufficient information state contains the phase-conditioned belief of every component. A single observation-based policy is optimized outside the phase mixture. The Gaussian coexistence statement is made for the correctly centered and scaled random variable.

1.1. A genuine latent-phase experiment. Fix a finite family of continuous order parameters $\Xi_\varepsilon : \Omega_\varepsilon \rightarrow \mathbb{R}^m$. In the Sinai case these are empirical recurrent-component occupations and boundary-excursion mass; in the hard-sphere case they are the prepared density/contact order parameters used in the B1 saddle chart. The corresponding limiting rate has finitely many isolated minimizing compact sets $K_0, \dots, K_J$, separated by a positive distance. Choose disjoint Borel neighborhoods U_j with $K_j \Subset U_j$ and define

$$E_{\varepsilon,j} = \{\Xi_\varepsilon \in U_j\}, \quad E_{\varepsilon,\partial} = \Omega_\varepsilon \setminus \bigcup_{j=0}^J E_{\varepsilon,j}.$$

Let J_ε be the resulting label and put

$$w_{\varepsilon,j} = P_\varepsilon(E_{\varepsilon,j}), \quad P_{\varepsilon,j} = P_\varepsilon(\cdot \mid E_{\varepsilon,j}).$$

The marginal identity

$$P_\varepsilon = \sum_j w_{\varepsilon,j} P_{\varepsilon,j}$$

is then exact. The boundary label is retained as one of the finitely many components; it is not silently discarded.

Theorem 1.1 (Phase construction and common-speed component LDPs). *For sufficiently small fixed neighborhoods U_j , the following hold at the common microscopic speed a_ε .*

- (i) *Each nonempty $E_{\varepsilon,j}$ is a genuine positive event and the displayed mixture identity is exact.*
- (ii) *The weights have expansions*

$$w_{\varepsilon,j} = e^{-a_\varepsilon \alpha_j} a_\varepsilon^{-\gamma_j} (c_j + o(1)), \quad c_j > 0,$$

with $\alpha_j < \infty$ for every retained component.

- (iii) *Each conditional law $P_{\varepsilon,j}$ satisfies a good path LDP at speed a_ε with rate*

$$I_j(x) = I(x) - \inf_{U_j} I$$

on the closure of its phase cell and $+\infty$ off it.

- (iv) *The boundary phase obeys the same statement or has $\alpha_\partial > \min_j \alpha_j$; it is never assumed superexponentially small without proof.*

Proof. The events are disjoint Borel subsets of the original sample space, proving (i). Apply the A3 or B2 joint LDP to $(X_\varepsilon, \Xi_\varepsilon)$. The isolated-minimum assumption and the sharp A2 or B1 local coefficient theorem give the exponential costs α_j , polynomial dimensions γ_j , and positive Hessian/coarea constants c_j . The same inserted local theorem applied to a bounded path cylinder gives the conditional Laplace principle inside each cell. Exponential tightness upgrades it to the path LDP, proving (ii)–(iii). Apply the ordinary upper and lower bounds to the compact boundary between the U_j ; either it has its own minimizing stratum, in which case it is kept as a component, or compact separation gives a strictly larger rate. This proves (iv). $\square$

Theorem 1.2 (Exact labelled and marginal mixture LDP). *On the labelled space,*

$$\mathcal{I}(j, x) = \alpha_j - \alpha_* + I_j(x), \quad \alpha_* = \min_k \alpha_k.$$

The unlabelled marginal rate is

$$I_{\text{mix}}(x) = \min_j \{\alpha_j - \alpha_* + I_j(x)\}.$$

Both are good rates.

Proof. For a closed labelled set, sum finitely many component upper bounds and use that the largest exponential term determines the logarithmic asymptotic. For an open set, select one component and use its lower bound. The marginal formula is the contraction under $(j, x) \mapsto x$. A finite union of compact component sublevels is compact. $\square$

1.2. The sufficient phase-conditioned belief state. Let S_j be the hidden state space of phase j . Under phase j , C1 gives a belief $\pi_t^j \in \mathcal{P}(S_j)$ and a common dominated observation density $g_j^a(\pi_t^j, y)$. Define the information state

$$\mathbf{z}_t = (\rho_t(j), \pi_t^j : 0 \leq j \leq J), \quad \sum_j \rho_t(j) = 1.$$

Terms with $\rho_t(j) = 0$ use the C1 cemetery convention. For one common action a , update each component belief by its own Bayes map and update the phase weights by

$$\rho_{t+1}(j) = \frac{\rho_t(j) g_j^a(\pi_t^j, Y_{t+1})}{\sum_k \rho_t(k) g_k^a(\pi_t^k, Y_{t+1})}.$$

Theorem 1.3 (Sufficiency and common domination). *The state $\mathbf{z}_t$ is a controlled Markov sufficient statistic for the phase-mixture observation experiment. The likelihood increments are defined with respect to the common C1 stratified observation measure, are jointly measurable, and multiply under adaptive policies. In general neither the scalar posterior vector ρ_t nor a single common physical state is sufficient.*

Proof. Condition first on the latent label. C1 says that π_t^j is sufficient for future hidden transitions and observations in component j . Conditional on the observed history, the law is the finite mixture of these component predictive laws with weights $\rho_t(j)$. Bayes' formula gives the displayed update. Therefore the conditional law of the next information state depends only on $\mathbf{z}_t$ and the common action.

Every component density is relative to the same stratified measure u . The product of its one-step predictive densities is the component likelihood under an adaptive action because the action is measurable with respect to the past observation sigma-field. Hence likelihood ratios multiply. If two phases have different hidden kernels, their future laws depend on the separate π_t^j ; this proves the final assertion and explains why the larger state is necessary. $\square$

1.3. Shared-policy risk-sensitive dynamic programming. Let $\mathfrak{A}$ be the class of policies predictable from the common observation history. For a fixed policy α , write

$$Z_{t,j}^\alpha(\pi^j; h) = \mathbb{E}_j^{\alpha, \pi^j} \exp \left\{ \theta \left[\int_0^t c_j(\pi_s^j, a_s) ds + h_j(\pi_t^j) \right] \right\}.$$

The phase-blind value is

$$\mathcal{V}_t(\mathbf{z}; h) = \frac{1}{\theta} \log \sup_{\alpha \in \mathfrak{A}} \sum_j \rho(j) Z_{t,j}^\alpha(\pi^j; h).$$

The supremum is outside the sum. A different optimizer for each hidden phase is not permitted.

Theorem 1.4 (Posterior shared-control semigroup). *On the compact reachable information state, the operators $\mathbf{S}_t h = \mathcal{V}_t(\cdot; h)$ form a risk-sensitive dynamic-programming semigroup. In discrete time,*

$$e^{\theta \mathbf{S}_{n+1} h(\mathbf{z})} = \sup_{a \in \mathcal{A}(\mathbf{z})} e^{\theta c(\mathbf{z}, a)} \int e^{\theta \mathbf{S}_n h(\mathbf{z}')} \mathbf{P}^a(\mathbf{z}, d\mathbf{z}'),$$

and the continuous-time version is obtained from the B4 relaxed-control compactness. For a fixed common policy, but only for a fixed common policy,

$$e^{\theta V_t^\alpha(\mathbf{z}; h)} = \sum_j \rho(j) Z_{t,j}^\alpha(\pi^j; h).$$

Proof. Theorem 1.3 gives the controlled Markov property. Apply the tower property at an intermediate time and optimize over the first action and the common continuation policy. This yields the Bellman formula and semigroup property. Compactness and the C1 Feller theorem give a measurable maximizing selector. B4 supplies the continuous-time compact relaxed-control concatenation. The final log-sum identity is ordinary conditioning on the label before optimization; placing the supremum outside the sum preserves the information constraint. $\square$

1.4. Derived complex phase charts. For a phase cell $E_{\varepsilon,j}$ and complex source F , put

$$Z_{\varepsilon,j}(F) = \mathbb{E}[e^{a_\varepsilon F(X_\varepsilon)}; E_{\varepsilon,j}].$$

The A2/A4 or B1/B2 spectral/coefficient decompositions yield on a complex neighborhood

$$Z_{\varepsilon,j}(F) = e^{-a_\varepsilon \alpha_j} a_\varepsilon^{-\gamma_j} e^{a_\varepsilon q_j(F)} (c_j(F) + r_{\varepsilon,j}(F)),$$

where c_j is analytic and nonzero at zero and $r_{\varepsilon,j} \rightarrow 0$ locally uniformly.

Theorem 1.5 (Zero-free phase-pressure atlas). *There is a source neighborhood $\mathcal{U}_j$ and $\varepsilon_j > 0$ such that $Z_{\varepsilon,j}$ has no zero on $\mathcal{U}_j$ for $\varepsilon < \varepsilon_j$. The analytic branch*

$$q_{\varepsilon,j}(F) = a_\varepsilon^{-1} \log Z_{\varepsilon,j}(F) + \alpha_j + \frac{\gamma_j \log a_\varepsilon}{a_\varepsilon}$$

converges locally uniformly with all fixed derivatives to $q_j(F)$.

Proof. The A2/A4 leading Riesz projection or the B1 exact coefficient theorem gives the displayed expansion, including the positive real normalization $c_j(0) > 0$ and a locally uniform analytic remainder. Shrink $\mathcal{U}_j$ so that $|c_j(F)| \geq c_0$. Then $|r_{\varepsilon,j}(F)| < c_0/2$ for small ε , so the parenthesis is zero-free by the elementary Rouché inequality. The exponential prefactor is never zero. Analytic logarithms therefore exist, and Cauchy's formula gives locally uniform derivative convergence. The dominant-term estimate is an output of the upstream spectral/coefficient theorem, not an assumption of this theorem. $\square$

1.5. Correct second-order coexistence. Assume that in phase j a statistic TX_ε has center m_j and

$$\sqrt{a_\varepsilon}(TX_\varepsilon - m_j) \implies N(0, \Sigma_j).$$

Let

$$J_* = \{j : \alpha_j = \alpha_*\}, \quad \gamma_* = \min_{j \in J_*} \gamma_j.$$

Include in c_j the complete subexponential factor from the phase weight, local partition function, saddle determinant, and any declared conditioning window.

Theorem 1.6 (Labelled Gaussian coexistence with correct scaling). *Along every sequence for which the normalized constants converge, the labelled random variable*

$$(J_\varepsilon, \sqrt{a_\varepsilon}(TX_\varepsilon - m_{J_\varepsilon}))$$

converges to

$$\sum_{j \in J_{**}} \omega_j \delta_j \otimes N(0, \Sigma_j), \quad J_{**} = \{j \in J_* : \gamma_j = \gamma_*\},$$

where

$$\omega_j = \frac{c_j}{\sum_{k \in J_{**}} c_k}.$$

Equivalently, the label-aware shifted variable $m_{J_\varepsilon} + \sqrt{a_\varepsilon}(TX_\varepsilon - m_{J_\varepsilon})$ converges conditionally in phase j to $N(m_j, \Sigma_j)$. The unscaled variable TX_ε converges to the mixture of point masses $\sum_j \omega_j \delta_{m_j}$.

Proof. The weight expansion shows that phases with $\alpha_j > \alpha_*$ vanish exponentially and those with $\gamma_j > \gamma_*$ vanish polynomially among the exponential minimizers. The remaining normalized weights converge to the stated ω_j . Conditional characteristic functions converge to those of $N(0, \Sigma_j)$ by the component CLTs. Summing the finitely many terms proves the labelled convergence. The two equivalent statements follow by the continuous mapping theorem and make explicit which variable has mean m_j . $\square$

1.6. Phase-aware contractions. Let T be a C2 typed contraction that acts componentwise on the phase path spaces and maps the phase-conditioned information state continuously. A conditioning sequence is admissible when the A2 or B1 inserted local theorem holds componentwise with the complete constants used above.

Theorem 1.7 (Phase contraction and tangent commutation). *The following operations commute under the stated verified interfaces:*

- (i) *the labelled mixture LDP followed by state contraction;*
- (ii) *component pressure followed by pullback of sources;*
- (iii) *component Gaussian tangent formation followed by the continuous linear tangent map;*
- (iv) *Bayes aggregation under one fixed common policy followed by optimization over common policies.*

The contracted rate is

$$I_T(j, y) = \alpha_j - \alpha_* + \inf_{Tx=y} I_j(x),$$

and the marginal rate is its minimum over j .

Proof. Item (i) is the C2 extended contraction principle applied separately to each of the finitely many labelled components, followed by the finite-mixture argument. The exact finite-volume source identity proves (ii), and locally uniform analytic convergence permits differentiation. The component CLT and the continuous linear mapping theorem prove (iii), including the phase-wise centering of Theorem 1.6. For (iv), condition on the label under a fixed common policy, apply the exact log-sum identity, update the full phase-conditioned belief state, and only then take the supremum. The Bellman proof in Theorem 1.4 shows that this order is stable under concatenation. $\square$

1.7. Mechanical and valuation parameters. For the A1 work cocycle W_n and an entropic valuation parameter ϑ , the normalized likelihoods agree for all paths only if the explicit likelihood-cocycle compatibility axiom is imposed. Under that axiom, nonconstancy of W_n forces equality of the two coefficients. Without it, mechanical work and valuation remain distinct typed coordinates. This separation is preserved by every contraction above.

Theorem 1.8 (Round-twenty-one D1 closure). *The phase synthesis is based on an exact finite-volume latent partition, component LDPs and complete weight asymptotics, a sufficient state of phase-conditioned beliefs, common likelihood domination, a shared-policy risk-sensitive DPP, derived zero-free complex pressure charts, a correctly scaled labelled Gaussian coexistence theorem, and verified phase-aware contractions.*

Proof. Combine Theorems 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, and 1.7. □

SCOPE AND VERIFICATION STATUS

This manuscript states a positive theorem reconstruction in the model-specific regular regime declared in its controlling theorems. Cross-paper inputs follow the Round-Twenty-One dependency ledger. Repository source identity, proof-structure, counterexample, and build checks are internal reproducibility gates and do not replace independent external mathematical review.

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